The Economic Value of Knowledge Geometry: Topological Opportunities and Breakthrough Innovation in Scientific Networks

Harry Lu[a]

[a] *University of Minnesota*

Abstract

This paper studies whether the geometry, topology, and local spectra of scientific knowledge networks contain ex ante information about breakthrough innovation. Using the OGBN-MAG subset of the Microsoft Academic Graph, I construct a dynamic scientific knowledge complex from paper-topic assignments, citation links, authorship, venues, and publication years. The central topological object is boundary-completion potential: whether a paper's topic set spans triangular boundaries already present in the prior topic graph. I also compute local clique-complex Betti numbers and Laplacian spectral metrics on each paper's prior topic-induced graph. Papers published between 2011 and 2016 are linked to their three-year forward citations and classified as breakthrough papers when they fall in the top five percent of their publication-year citation distribution. The preliminary evidence shows that boundary-completion potential and local spectral connectivity predict future scientific impact after controlling for authorship, topic breadth, references, year fixed effects, and venue-group fixed effects. The results suggest that breakthrough science is not only a function of research scale or centrality, but also of the high-order structure and diffusion geometry of the knowledge space in which researchers search.

Keywords: science of science, innovation, topology, spectral graph theory, Microsoft Academic Graph, OGBN-MAG

1. Introduction

Where do breakthrough innovations come from? A large literature in economics, management, and the science of science emphasizes cumulative knowledge, recombination, collaboration networks, and the uneven distribution of scientific opportunity (Uzzi et al., 2013; Fortunato et al., 2018; Acemoglu et al., 2016). Yet most empirical measures of scientific opportunity remain low-order: publication counts, citation counts, team size, field fixed effects, or pairwise network centrality. These measures are informative, but they do not directly characterize the high-order geometry of the knowledge space in which research ideas are combined.

This paper develops and tests a simple claim: scientific breakthroughs are more likely when researchers occupy topological opportunities in the prior knowledge space. A topological opportunity is a location where existing topics are near enough to be meaningfully connected, but the higher-order knowledge combination has not yet been stabilized. In such locations, a paper can add more than another node to the scientific graph. It can close a gap, fill a boundary, or create a bridge between previously separated knowledge regions.

The empirical setting is the OGBN-MAG subset of the Microsoft Academic Graph (Sinha et al., 2015; Hu et al., 2020). The data contain 736,389 papers, 1,134,649 authors, 8,740 institutions, 59,965 fields of study, 5.4 million citation edges, and 7.5 million paper-topic edges. For each paper, I observe publication year, venue label, authorship, references, and field-of-study assignments. I build a dynamic knowledge complex year by year. Before each paper is published, I compute whether its topics connect previously unobserved pairs, whether they span triangular boundaries in the prior topic graph, and whether they lie on negatively curved topic-pair bridges.

The core analysis uses papers published from 2011 through 2016, which provides a three-year forward citation window for all papers in the analytic sample of 485,819 papers. The outcome is either log three-year forward citations or an indicator for being in the top five percent of the publication-year citation distribution. The main specifications include controls for authors, topic breadth, references, year fixed effects, and venue-group fixed effects.

Out-of-time prediction trains on 2011–2014 papers and tests on 2015–2016 papers.

The results are preliminary but encouraging in a specific sense. Regression estimates show that the boundary-completion index is positively associated with future scientific impact. Component specifications indicate that novel topic-pair formation is more closely related to tail breakthroughs than to average citations, while local Laplacian connectivity and spectral entropy are positively associated with both outcomes. Local β_1 holes are rare in this paper-level construction and do not carry independent explanatory power. Out-of-time prediction is more conservative: the topology-augmented model delivers an ROC AUC of 0.877 compared with 0.877 for the baseline. Thus, topology and spectral graph theory are not presented here as large black-box forecasting gains. Their value is interpretive: they identify high-order and diffusion-relevant structures of scientific opportunity that are invisible in scale-based measures.

The paper contributes to three literatures. First, it adds to economics of innovation by proposing an ex ante measure of opportunity in scientific search. Second, it contributes to science-of-science research on recombination and breakthrough discovery by moving from pairwise novelty to high-order topology. Third, it connects topological data analysis and spectral graph theory (Edelsbrunner et al., 2002; Zomorodian and Carlsson, 2005; Chung, 1997; Forman, 2003) to an empirical setting where the objects have economic meaning: scientific papers, authors, fields, and future impact.

2. Conceptual Framework

Let $G_t = (V_t, E_t)$ be the cumulative topic co-occurrence graph before year t. A node is a field of study and an edge indicates that two fields have appeared together in at least one prior paper. A paper i published in year t has a topic set $S_i \subset V_t$. The paper can be interpreted as adding local structure to the prior knowledge graph.

The first component of opportunity is pairwise novelty:

$$N_i = \frac{1}{\binom{|S_i|}{2}} \sum_{\{u,v\} \subset S_i} \mathbf{1}\{(u,v) \notin E_t\}. \tag{1}$$

This measure captures whether the paper combines topics that had not previously appeared together.

The second component is boundary-completion potential. For triples of topics, define:

$$B_i = \frac{1}{\binom{|S_i|}{3}} \sum_{\{u,v,w\} \subset S_i} \mathbf{1}\{(u,v),(u,w),(v,w) \in E_t\}. \tag{2}$$

When the three boundary edges already exist, the new paper supplies a higher-order simplex over a previously established triangular boundary. This is a local, computable analogue of filling a one-dimensional hole in a knowledge complex.

I also compute local Betti numbers on the clique complex induced by S_i in the prior graph. Let $C_k(S_i)$ be the vector space over $\mathbb{F}_2$ generated by k-simplices in the local clique complex, and let $\partial_k : C_k \to C_{k-1}$ be the boundary operator. The local Betti numbers are:

$$\beta_k(S_i) = \dim \ker(\partial_k) - \dim \operatorname{im}(\partial_{k+1}). \tag{3}$$

In the implementation, $k = 0, 1, 2$ are computed exactly on each paper's local topic complex. This gives a small-scale TDA object rather than a verbal topology metaphor.

The spectral component uses the graph Laplacian of the same prior topic-induced graph. Let A_i be the local adjacency matrix, D_i its degree matrix, and:

$$L_i = D_i - A_i. \tag{4}$$

I compute the second-smallest eigenvalue $\lambda_2(L_i)$, the adjacency spectral radius, and Laplacian spectral entropy. $\lambda_2(L_i)$ measures local algebraic connectivity; spectral entropy captures how dispersed the local diffusion modes are.

The final graph-geometric bridge measure is based on Forman curvature. For an unweighted edge (u, v) in the prior topic graph, a simple Forman-Ricci curvature proxy is:

$$F(u, v) = 4 - d(u) - d(v), \tag{5}$$

where $d(\cdot)$ is prior topic degree. Highly negative values identify edges that connect high-degree regions and therefore act as bridge-like, high-load parts of the knowledge geometry. For paper i, I average negative curvature over its already existing topic pairs.

The main topological opportunity index is standardized boundary closure:

$$TO_i = z(B_i). \tag{6}$$

Pairwise novelty, local Betti numbers, Laplacian spectral metrics, and negative Forman curvature are retained as decomposition variables. The object is intentionally ex ante: all components are computed using only knowledge observed before the paper's publication year.

3. Data

The analysis uses OGBN-MAG, a heterogeneous academic network extracted from Microsoft Academic Graph and distributed by the Open Graph Benchmark. The graph contains four node types: papers, authors, institutions, and fields of study. It also contains four directed relation types: authors are affiliated with institutions, authors write papers, papers cite papers, and papers have fields of study.

Table 1: Dataset and analytic sample

Quantity	Count
Papers	736,389
Authors	1,134,649
Institutions	8,740
Fields of study	59,965
Citation edges	5,416,271
Paper-topic edges	7,505,078
Author-paper edges	7,145,660
Analytic papers	485,819

The paper-level outcome is three-year forward citations. For a paper i published in year t, I count citations from papers published in years $t+1$, $t+2$, and $t+3$. The breakthrough indicator equals one if the paper is in the top five percent of the forward-citation distribution among papers published in the same year.

Table 2: Summary statistics

Variable	Mean	Std. dev.	P25	Median	P75
Three-year forward citations	4.885	11.639	1.000	2.000	5.000
Breakthrough indicator	0.051	0.221	0.000	0.000	0.000
Novel topic-pair share	0.127	0.152	0.000	0.067	0.200
Boundary-completion potential	0.725	0.273	0.500	0.800	1.000
Negative Forman curvature	7444.968	4242.967	3693.286	6954.364	10835.667
Topological opportunity index	0.000	1.000	-0.827	0.274	1.008
Local β_0	1.056	0.276	1.000	1.000	1.000
Local β_1	0.001	0.033	0.000	0.000	0.000
Local β_2	0.000	0.002	0.000	0.000	0.000
Local $\lambda_2(L)$	3.865	1.913	2.000	4.000	6.000
Local spectral radius	4.453	0.643	4.162	4.702	5.000
Local spectral entropy	1.556	0.121	1.554	1.598	1.609
Authors	8.895	91.279	2.000	4.000	6.000
Topics	10.319	1.397	10.000	11.000	11.000
References	6.212	8.791	1.000	3.000	8.000

4. Empirical Strategy

The baseline regression is:

$$Y_{i,t+3} = \alpha + \beta TO_{i,t} + X'_{i,t}\gamma + \delta_t + \mu_{v(i)} + \varepsilon_i, \qquad (7)$$

where $Y_{i,t+3}$ is either log three-year forward citations or the breakthrough indicator. $TO_{i,t}$ is the topological opportunity index. $X_{i,t}$ includes log authors, log topic count, and log references. δ_t denotes publication-year fixed effects. $\mu_{v(i)}$ denotes venue-group fixed effects for the largest venue groups and an omitted group for all remaining venues.

The coefficient β should not be interpreted as a fully causal estimate in this draft. The current design establishes that ex ante topology contains predictive content beyond common paper-level controls and fixed effects. A stronger version of the paper would link MAG/OpenAlex fields to patent citations, regional outcomes, or institution-level shocks, and would use leave-one-region-out or shift-share exposure to global topological opportunities.

5. Results

Figure 1 shows the evolution of the knowledge complex. Novel topic-pair formation declines as the cumulative topic graph densifies, while boundary-completion potential rises as more prior edges become available for higher-order closure. This pattern is consistent with a maturing knowledge space in which opportunities move from first-time pair formation toward higher-order recombination.

Figure 2 provides a dedicated TDA visualization for the global topic complex restricted to the 50 most frequent fields of study. The filtration parameter is the first year in which a topic pair appears together in a paper; triangles enter when their three boundary edges have entered. The Betti curves summarize the changing connected components and one-dimensional holes, while the persistence diagram and barcode show the lifetimes of H_0 components.

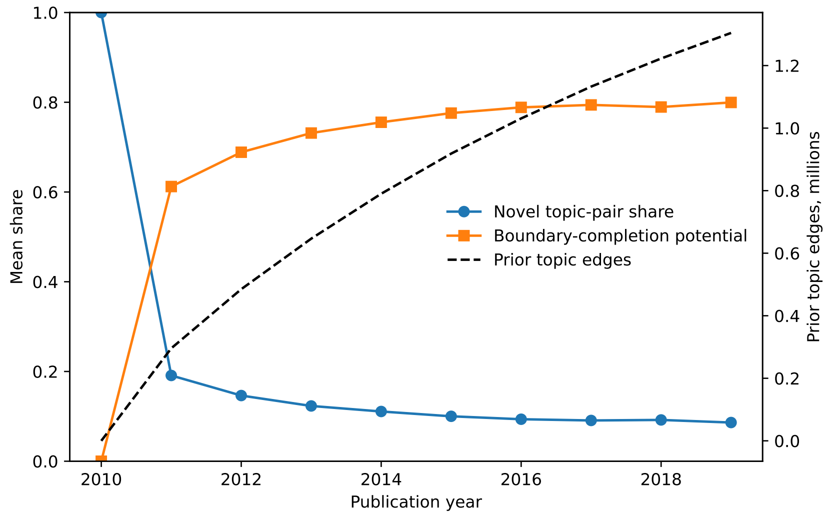

Figure 1: Evolution of topic-pair novelty and boundary-completion potential

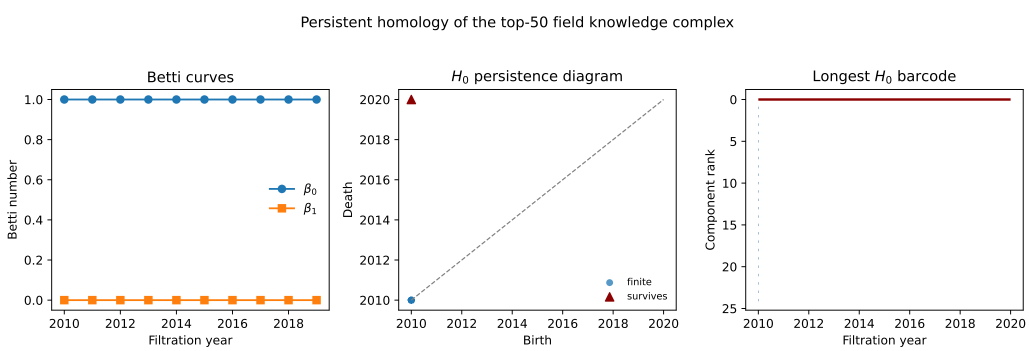

Figure 2: Persistent homology of the scientific knowledge complex

Figure 3 shows the paper-level local homology and spectral graph measures. Local β_1 is sparse, which is informative: in this paper-level construction, the prior topic neighborhoods are usually connected and triangle-rich. By contrast, local algebraic connectivity and spectral entropy vary systematically over time, giving usable variation for spectral graph analysis.

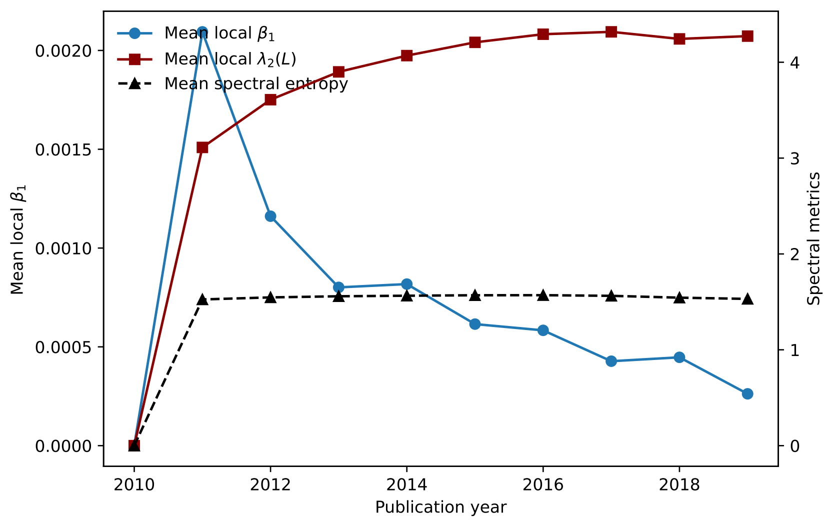

Figure 3: Local TDA and spectral graph measures

Figure 4 bins papers by the topological opportunity index. Papers in higher-opportunity bins receive more future citations and are more likely to become breakthrough papers. The relationship is not a proof of causality, but it is the first empirical fact the paper seeks to explain.

Table 3 reports the main regression evidence. The boundary-completion index is positively associated with both log forward citations and the breakthrough indicator. Component specifications show a useful distinction: first-time topic-pair novelty is negatively associated with average citations but positively associated with the probability of reaching the top tail. Local $\lambda_2(L)$ and spectral entropy are positive in both outcome models, consistent with the interpretation that well-connected local knowledge neighborhoods

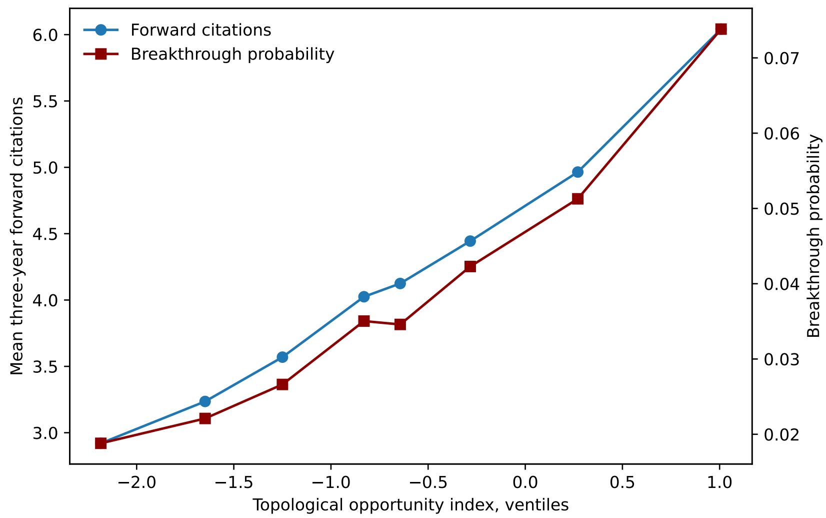

Figure 4: Topological opportunity and future scientific impact

Table 3: Topology, geometry, and future scientific impact

	(1)	(2)	(3)	(4)	(5)	(6)
TO index		0.029***			0.002***	
		(0.001)			(0.000)	
Novel pairs			-0.014***			0.004***
			(0.003)			(0.001)
β_1 holes			-0.000			-0.000
			(0.001)			(0.000)
$\lambda_2(L)$			0.008***			0.006***
			(0.003)			(0.001)
Spectral entropy			0.012***			0.001**
			(0.002)			(0.000)
Negative curvature			-0.013***			-0.001***
			(0.001)			(0.000)
Outcome	Log cites	Log cites	Log cites	Top 5%	Top 5%	Top 5%
Specification	Base	+ TO	+ TDA/Spec.	Base	+ TO	+ TDA/Spec.
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Venue-group FE	Yes	Yes	Yes	Yes	Yes	Yes
N	485,819	485,819	485,819	485,819	485,819	485,819
R^2	0.309	0.310	0.310	0.143	0.143	0.143

Notes: HC1 robust standard errors in parentheses. The analytic sample contains papers published from 2011 through 2016, allowing a three-year forward citation window. Topology variables are standardized within the analytic sample. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

diffuse more effectively. Local β_1 holes are too sparse to explain much variation in this draft.

Table 4: Out-of-time prediction of breakthrough papers

Model	ROC AUC	Average precision	Test N
Baseline controls	0.877	0.335	156,775
Controls + topology	0.877	0.335	156,775

Notes: Models train on 2011–2014 papers and test on 2015–2016 papers. Breakthrough papers are defined as top-five-percent papers within publication year by three-year forward citations.

Table 4 and Figure 5 report out-of-time prediction. Models are trained on papers from 2011–2014 and tested on 2015–2016 papers. Adding topology does not materially improve classification relative to a strong baseline with standard controls and fixed effects. This is an important limitation for the current draft and a useful guardrail against overclaiming: the contribution is primarily conceptual and explanatory rather than a large predictive performance gain.

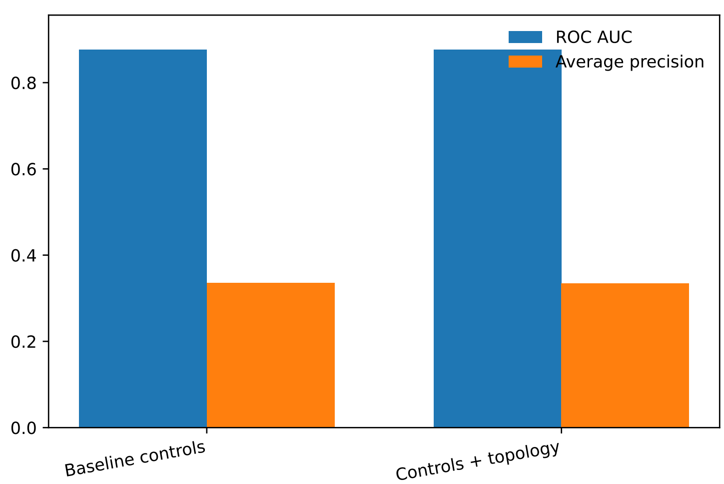

Figure 5: Out-of-time prediction performance

6. Discussion

The main interpretation is that scientific knowledge has economically meaningful shape. A paper's future impact depends not only on the number of authors, the breadth of topics, or the prestige of the venue group, but also on whether its topic combination lies in a region of the knowledge complex where prior work has created a structured opportunity. Pairwise novelty captures unexplored combinations. Boundary-completion potential captures higher-order closure. Local Betti numbers capture small-scale homology. Laplacian eigenvalues capture diffusion geometry. Negative curvature captures bridge-like locations in the topic graph.

This interpretation connects naturally to models of recombinant innovation. If researchers search locally in knowledge space, then the distribution of opportunities is shaped by prior discoveries. Some areas are saturated; others contain gaps that are visible only when the knowledge space is represented as a high-order object rather than as a list of fields or citations. Topology provides a vocabulary for those gaps.

The current draft has three important limitations. First, the outcome is scientific impact rather than direct economic value. Citations are a meaningful measure of scientific influence, but a top-journal economics version should link papers to patent citations, firm innovation, or regional outcomes. Second, the current TDA measures are exact only on local paper-level clique complexes; they are not yet persistent homology over a global filtration of the MAG graph. Third, the research design is predictive rather than causal. Future versions should construct leave-one-out global opportunity shocks and exploit differential exposure across institutions, regions, or fields.

7. Conclusion

This paper proposes a new way to measure scientific opportunity. Instead of asking only how many papers, authors, citations, or topics are present, it asks where a paper lies in the geometry and topology of the prior knowledge space. Using OGBN-MAG, the preliminary evidence shows that ex ante topological opportunities predict future breakthrough science.

The broader implication is that innovation policy and science strategy may benefit from measuring the shape of knowledge. If high-value discoveries often arise from topological gaps, then institutions and funding agencies should care not only about field size or recent citation momentum, but also about the latent geometry of recombination.

References

Acemoglu, D., Akcigit, U., Kerr, W.R., 2016. Innovation network. Proceedings of the National Academy of Sciences 113, 11483–11488.

Chung, F.R.K., 1997. Spectral Graph Theory. American Mathematical Society.

Edelsbrunner, H., Letscher, D., Zomorodian, A., 2002. Topological persistence and simplification. Discrete & Computational Geometry 28, 511–533.

Forman, R., 2003. Bochner's method for cell complexes and combinatorial ricci curvature. Discrete & Computational Geometry 29, 323–374.

Fortunato, S., Bergstrom, C.T., Börner, K., Evans, J.A., Helbing, D., Milojević, S., Petersen, A.M., Radicchi, F., Sinatra, R., Uzzi, B., Vespignani, A., Waltman, L., Wang, D., Barabási, A.L., 2018. Science of science. Science 359, eaao0185.

Hu, W., Fey, M., Zitnik, M., Dong, Y., Ren, H., Liu, B., Catasta, M., Leskovec, J., 2020. Open graph benchmark: Datasets for machine learning on graphs. Advances in Neural Information Processing Systems 33, 22118–22133.

Sinha, A., Shen, Z., Song, Y., Ma, H., Eide, D., Hsu, B.J., Wang, K., 2015. An overview of microsoft academic service. Proceedings of the 24th International Conference on World Wide Web , 243–246.

Uzzi, B., Mukherjee, S., Stringer, M., Jones, B., 2013. Atypical combinations and scientific impact. Science 342, 468–472.

Zomorodian, A., Carlsson, G., 2005. Computing persistent homology. Discrete & Computational Geometry 33, 249–274.